

signal to each other. By the end of the 19th century, the neuron doctrine had replaced reticular theory. It gained widespread acceptance in the 1950s, when electron microscope images revealed the existence of synapses. Today, it is the model for the structure and function of the nervous system.

In 1906, Cajal and Golgi were jointly awarded the Nobel Prize in Medicine or Physiology for their separate studies on the nervous system; Golgi, however, was a staunch reticulist and still insisted that nerve cells were physically connected.

“As long as our **brain** is a **mystery**, the **Universe**, the reflection of the structure of the brain, **will also be a mystery.**”

Santiago Ramón y Cajal, 1920



CAMILLO GOLGI

Italian doctor Camillo Golgi invented a tissue-staining technique.



While studying the nervous system, Golgi (1843–1926) wanted to make the structure of nerve cells clearer under a microscope. In 1873, he discovered how to make them show up by staining them black with silver nitrate, a method Cajal greatly improved. Despite mounting evidence that he was wrong, Golgi clung tenaciously to the accepted reticular theory and even argued against Cajal's ideas when the two men collected their joint Nobel Prize.

Cajal used pen and ink to reproduce, with almost photographic precision, the intricate structure of the brain and nervous system. His drawings were a powerful tool for transmitting his observations to the scientific world.